CariSurg MedTech Pathways Research Project

Week 5 - Feasibility Memo Outline

Project Title: Enhancing Emergency Department Triage with Artificial Intelligence within the Caribbean Context

Student Name: Imani Zakuri

Student GitHub: https://github.com/imaniiz/CariSurg-Portfolio

Purpose: To asssess whether the emergency department dataset is suitable for developing a baseline AI-assisted triage model and identify any limitations that should be addressed before modelling.

Memo Outline:

Section 1: Verdict:
Provide a one-sentence recommendation on whether the dataset is suitable for building a baseline triage model. Include overall recommendation and brief justifications

Section 2: Dataset Summary
Provide a brief overview of the dataset including the following information:

1. Number of patient encounters
2. Number of clinical features
3. Structured variables and chief complaint flags
4. Target variable (Emergency Severity Index - ESI)
5. Patient demographics (age, gender, race, ethnicity)
6. Distribution of triage levels
7. Supporting figures such as ESI and demographic distributions

Section 3: Top 3 Data Quality Concerns

Concern 1: Missing Data

- Extent of missing values
- Features most affected
- Proposed handling approach

Concern 2: Outcome Leakage

- Variables unavailable at triage (e.g., disposition)

- Reason these variables should be excluded from modelling

Concern 3: Representativeness

- Dataset collected from a single U.S. hospital
- Potential limitations when applying findings to Caribbean emergency departments

Supporting figures:

- Missing values table or heatmap

Section 4: Top 3 Reasons to Process

Explain why the dataset is still appropriate for developing a baseline model. Include discussion points such as:

- Availability of real ESI labels
- Rich collection of demographic and clinical variables
- Comprehensive triage vital signs
- Large number of patient encounters
- Well-documented and reproducible data cleaning process
- Supporting figures such as vital sign distributions and chief complaint frequencies

Section 5: Caveats

Discuss limitations that should be acknowledged before model development. Potential caveats include:

- Temperature recorded in Fahrenheit rather than Celsius
- Sparse chief complaint variables
- Correlation does not imply predictive value
- External validation required before deployment in Caribbean healthcare settings
- Additional evaluation and modelling will be completed in Week 6

Section 6: Top 10 Candidate Predictive Features

Briefly introduce the features that will be investigated in Week 6 as potential predictors of triage level. Potential candidates include:

- Age
- Heart rate
- Respiratory rate
- Blood pressure
- Oxygen saturation
- Temperature
- Arrival mode

- Selected chief complaint indicators
- Gender
- Race and ethnicity (primarily to assess fairness and bias rather than predictive performance)